INTERNATIONAL TELECOMMUNICATION UNION

TELECOMMUNICATION STANDARDIZATION SECTOR

STUDY PERIOD 2022-2024

SG9-TD162

STUDY GROUP 9

Original: English

Question(s): 3, 5, 8, 9, 11/9 E-meeting, 6-14 September 2022

TD

Source: Chairman WP2/9

Title: Report of Working Party 2/9 meeting

Contact:	TaeKyoon KIM ETRI Korea (Rep. of)	Tel: Fax: Email:	82-42-860-6917 82-42-860-6465 tkkim@etri.re.kr

Abstract: This TD contains the meeting report for Working Party 2/9 meeting.

Table of Contents

1 Introduction

Working Party 2/9 "Cable-related platforms and applications" met during the SG9 meeting held fully virtual on 6-14 September 2022, chaired by Mr Taekyoon Kim (ETRI, Rep. of Korea), and assisted by Mr Stefano Polidori (TSB Advisor).

The meeting was addressed in one session on 13 September 2022. The group adopted the agenda in SG9-TD191-R1.

The structure and leadership of WP2/9 is reproduced in ANNEX A.

The list of participants is reproduced in ANNEX B.

2 Objectives

The objective of WP2/9 meeting is to discuss, during the closing Plenary of WP2/9, the activities of each Question under WP2/9, as they are reported by each Rapporteur. This normally includes:

– draft Questions reports
– draft Recommendations for AAP consent
– draft Recommendations for progressing
– proposed new/deleted work items
– outgoing liaison statements
– future meetings
– appointment of Rapporteurs/Associate/liaison officers
– any other business

3 Documentation and email lists

3.1 Documentation

The following documents were discussed by WP2/9 Questions:

Q3/9: C: 40R1. TD: 31, 38, 66,71, 72, 74, 87, 96, 113, 115, 116, 122, 148

Q5/9: TD: 38, 59, 60, 65, 71, 106, 128

Q8/9: C: 23, 24, 25, 26. TD: 20, 21, 22, 38, 39, 40, 65, 71, 91, 96, 100, 114

Q9/9: C: 29,38,41,42. TD: 38, 53, 65, 66, 71, 75, 87, 95, 96, 98, 103, 104, 105, 107, 108, 109, 148

Q11/9: C: 35, 36. TD: 57, 58, 71, 93, 101, 153, 154

The complete documentation for this SG9 meeting is to be found at:

https://www.itu.int/md/T22-SG09-220906/sum/en

3.2 Emailing list subscription

E-mail correspondences pertaining to the activities of this group are routinely conducted using the e-mail reflector t22sg9all@lists.itu.int. Those wishing to subscribe or unsubscribe to this or other SG9 email reflectors should visit the mailing list web page at:

https://www.itu.int/net4/iwm?p0=0&p11=ITU-SEP-ITU-T&p12=ITU-SEP-ITU-T-SEP-SP%2017-SEP-Study%20Group%209&p21=ITU&p22=ITU

Information on SG9 informal FTP areas and all SG9 mailing lists is on the related webpage .

4 WP2/9 meeting results

The reports of each Question under WP2/9 were reviewed and approved. The executive summary of the Questions meetings results is reported below. Reference to the Questions reports is provided for more details.

4.1 Question 3/9

The report of Q3/9 can be found in SG9-TD133-R1. It was approved.

Question 3 meet in two sessions and reviewed one contribution and 12 incoming liaisons.

In particular, Q3/9 reviewed the contribution C40R1, which describes the E2E network characteristics requirement for video services. SG9 management team and some members from Q3 recommended to update the scope and the title, in addition to its overall content, to avoid the overlap of scope with ITU-T SG12. Some members (e.g. Huawei, Ukraine) from Q3/9 committed to submit contributions to advance ITU-T J.pcnp-char (planned J.1630).

Q3/9 reviewed the incoming liaison TD38, which requests to update the Supplement ITU-T Y.sup.aisr (SG13-TD834/WP2), Supplement on Artificial Intelligence Standardization Roadmap. Q3 will collect the NWIs from SG9 and assess whether is necessary to update the supplement. At this meeting, an outgoing liaison statement to SG13, to update the Supplement, has been prepared by Q10/9.

Q3/9 drafted one outgoing liaison statement to ITU-T SG12, in collaboration with Q9.

Q3/9 planned one interim meeting on 24 November 2022.

4.2 Question 5/9

The report of Q5/9 can be found in SG9-TD135-R1. It was approved.

Q5/9 addressed five incoming liaison statements and one meeting report during its meetings.

The group reviewed the five incoming liaison statements, which were noted.

The group produced one outgoing liaison statement as a reply to SG16-LS282 on smart TV Operating System. The outgoing LS is available in TD172-R1.

Q5/9 did not plan interim meetings until next SG9 meeting in 2023.

4.3 Question 8/9

The report of Q8/9 can be found in SG9-TD138-R3. It was approved.

Q8/9 achieved the following results:

– Two new work items were agreed to be initiated:
    - draft Recommendation ITU-T J.mma-req on “Requirements of microservice architecture for audio-visual media in the converged media cloud”, the output is available in TD163.

  - o draft new Recommendation ITU-T J.mma-spec on "Specification of microservice architecture for audio-visual media in the converged media cloud", the output is available in TD164.

- One outgoing LS on the initiation of new work item ITU-T J.mma-req and J.mma-spec was agreed to be sent to ITU-T SG13 and ITU-T SG16. It's available in TD160.

One interim meeting is planned: 23 November 2022, e-meeting.

ACTION FOR SG9 PLENARY:
The date of consent for ITU-T J.mma-req and ITU-T J.mma-spec was not agreeable during Q8/9 sessions, nor WP2 plenary. SG9 Plenary is requested to discuss further.

- ***Option1: propose these two draft Recommendations for consent at next SG9 meeting in 2023.***
- ***Option2: propose these two draft Recommendations for consent at a WP2 meeting in 2023 (before next SG9 meeting).***
- ***Option3: propose these two draft Recommendations for consent at a WP2 meeting in December 2022***

4.4 Question 9/9

The report of Q9/9 can be found in SG9-TD139-R1. It was approved.

The Q9/9 sessions operated under WP2, under the chairmanship of Rapporteur Eric Wang, assisted by the Associate Rapporteur Soonchoul Kim.

Q9/9 covers requirements, methods, and interfaces of the advanced service platforms to enhance the delivery of audiovisual content, and other multimedia interactive services over integrated broadband cable networks.

Q9/9 met for three (3) sessions with about 28 people attending and addressed four (4) contributions and eight (8) incoming liaison statements.

Q9/9 decided not to proceed with AAP consent for J.cable-mabr this time, even if it was planned for consent, because of an unresolved issues that needs a feedback from DVB. An outgoing liaison statement was issued in this regards.

Q9/9 progressed the J.cloud-vr-arch and J.cloud-game-req.

Q9/9 used the "issues list" tool for maintaining the work of Q9/9. The approved "Q9 issues list" is found in TD175.

Q9/9 produced two (2) outgoing liaison statements, available in TD168-R1 and TD171-R1.

4.5 Question 11/9

The report of Q11/9 can be found in SG9-TD141. It was approved.

Q11/9 achieved the following results:

- One new work item was agreed:
  - o Draft ITU-T Technical Report JSTR.LCAP "Technical advances, challenges, and best practices in live captioning", the output is available in TD159
- One interim meeting is planned: Between 23 Nov and 07 December 2022 (TBD), E-meeting.
- One outgoing liaison statement to ITU-R SG6, ITU-T SG16, and IRG-AVA was approved, available in TD166.

New Associate Rapporteur was nominated: Mr. Zhao Ming, China.

5 Draft new/revised Recommendations proposed for "consent" (AAP) and/or "determination" (TAP)

The following draft Recommendations are proposed for consent/determination:

#	Question	Rec	Status	Title	Final TD GEN	A.5 justification
	none					

NOTE1: According to the decision of TSAG, at its meeting on 8-11 February 2010, the Rapporteur ensures that all of the bullet points of the "Check list to draft Recommendations" have been reviewed and that they have been fulfilled adequately by the above draft Recommendation(s).

Also the Rapporteur checked that the editor had applied the skeleton template to draft Recommendations and that this Recommendation is compliant with the Author's Guide.

6 Supplements, Technical Reports or other documents for agreement/approval

The following documents are proposed for agreement/approval:

#	Question	Kind of publication	Status	Title	Final TD GEN
		None			

7 Intellectual property statements

No IPR statements were received at this meeting.

8 Outgoing liaison statements

The following LSs were reviewed and approved by the WP2/9:

#	Questions	WP	To	For	Title	TD
1	3/9, 9/9	2/9	ITU-T SG12, ITU-T SG16, ETSI ISG F5G	Information	LS/o on ITU-T J.pcnp-char: "E2E Network Characteristics Requirement for Video Services", J.cloud-game-req: "Requirements of E2E Network Platform for Cloud Gaming Service", and J.cloud-vr-arch: "Architecture of E2E Network Platform for Cloud-VR Service"	TD171-R1
2	Q5/9	WP2	TSAG RG-WM ITU-T SG16	Action Information	LS/o/r on smart TV Operating System (SG16-LS282)	TD172-R1
3	Q8/9	WP2	ITU-T SG13, ITU-T SG16	Information	LS/o on the initiation of new work item ITU-T J.mma-req and J.mma-spec	TD160
4	Q9/9	2/9	DVB TM-TCAST, ITU-T SG16, CableLabs	Action Information	LS/o on the final baseline text for J.cable-mabr "Requirements of multicast adaptive bitrate (M-ABR) IP delivery."	TD168-R1
5	Q11/9	WP2	ITU-R SG6; ITU-T SG16 Q26/16, IRG-AVA	Information	LS/o to inform the decision of SG9 to initiate a new work item on proposed new ITU-T Technical Report JSTR.LCAP "Technical advances, challenges, and best practices in live captioning"	TD166

9 Work programme

9.1 New work items

The meeting agreed to start work on the following new work items:

#	Q.	Acronym (kind of publication)	Status	Title	Editor	TD	A.1 Justification template
1	Q8/9	J.mma-req (Draft Rec.)	New	Requirements of microservice architecture for audio-visual media in the converged media cloud	Mr Jia Tao (jiatao@abp2003.cn)	TD163	ANNEX C (A.1)
2	Q8/9	J.mma-spec (Draft Rec.)	New	Specification of microservice architecture for audio-visual media in the converged media cloud	Mr Jia Tao (jiatao@abp2003.cn)	TD164	ANNEX D (A.1)
3	Q11/9	JSTR.LCAP (Technical Report)	New	Technical advances, challenges, and best practices in live captioning	Avinash Agarwal, Ministry of Communications, India	TD159	ANNEX E (A.13)

					(avinash.70@gov.in) Rajiv Khattar, India (rajivkhattar@gmail.com)		

9.2 Deleted work items

The meeting agreed to stop the work on the following work items:

#	Q.	Acronym (kind of publication)	Title
1		None	

9.3 Total ongoing work items for progressing at next meeting

The current list of open work items, including eventual new work items agreed at this meeting (see item 9.1 above) are as follows:

#	Q.	Acronym (kind of publication)	Status	Title	Editor	TD	Time Consent
1	Q3/9	J.pcnp-char (planned J.1630)	New	E2E Network Characteristics Requirement for Video Services	Yanbin (Evan) Sun	TD165	2023
2	Q8/9	J.mma-req (Draft Rec.)	New	Requirements of microservice architecture for audio-visual media in the converged media cloud	Mr Jia Tao (jiatao@abp2003.cn)	TD163	2023
3	Q8/9	J.mma-spec (Draft Rec.)	New	Specification of microservice architecture for audio-visual media in the converged media cloud	Mr Jia Tao (jiatao@abp2003.cn)	TD164	2023
4	Q9	J.cable-mabr	New	Requirements of multicast assisted adaptive bitrate (M-ABR) IP delivery	Mr. Tatsuo Shibata (t-shibata@jlabs.or.jp)	TD167	2023-4
5	Q9/9	J.cloud-vr-arch	New	Architecture of E2E Network Platform for Cloud-VR Services	Mr. Evan Sun (evan.sun@huawei.com) Mr. Kei Kawamura (ki-kawamura@kddi.com)	TD169	2023-4
6	Q9/9	J.cloud-game-req	New	Requirements of E2E Network Platform for Cloud Gaming Services	Mr. Evan Sun (evan.sun@huawei.com)	TD170	2023-4

7	Q11/9	J.acc-us-prof (Draft Rec.)	New	Common user profile format for audiovisual content distribution	Pradipta Biswas (pradipta@iisc.ac.in)	TD153	2023
8	Q11/9	JSTR.LCAP (Technical Report)	New	Technical advances, challenges, and best practices in live captioning	Avinash Agarwal (avinash.70@gov.in) Rajiv Khattar (rajivkhattar@gmail.com)	TD159	2024

NOTE: The SG9 Work Programme can be found at: http://handle.itu.int/11.1002/db/wp-sg9 .

10 Planned interim Rapporteurs meetings

The following interim Rapporteurs' groups meetings for the Questions under WP2/9 are proposed for approval:

Question/SG	Date	Place / Host	Terms of reference	Contact
3/9	24th Nov 2022 (9h00-11h00 Geneva Time)	e-Meeting	Progress on J.1630 (Balyar Volodimir from Ukraine and Evan Sun from China will prepare the baseline text)	Evan Sun (evan.sun@huawei.com)
8/9	23 November 2022 (2 sessions from 11h30-14h30)	e-meeting	Progress on ongoing work items : J.mma-req, J.mma-spec. Discuss any additional contribution	Rakesh Desai (rakesh.desai@gov.in)
9/9	Date TBD, January 2023	e-meeting	Progress on J.cloud-vr-arch, J.cloud-game-req, J.cable-mabr, etc.	Eric Wang (eric.wangxiang@huawei.com)
11/9	Between 23 Nov and 07 December 2022 (TBD)	E-meeting (TBD)	Progress on J.acc-us-prof, JSTR.LCAP	Avinash Agarwal (avinash.70@gov.in)

Details will be posted on the ITU-T SG9 Rapporteur Group meetings website:
http://www.itu.int/net/ITU-T/lists/rgm.aspx?Group=9&type=interim

11 Scheduled WP or SG9 meetings

11.1 Interim WP2/9 meetings (before next SG9 meeting)

WP2/9 does not plan to organize a WP2/9 meeting, before the next SG9 meeting in March or April 2023 (TBD). However, WP2/9 may organize a meeting if SG9 Plenary decides on 14 September 2022 to resolve the issue related to Q8/9 draft Recommendations with a WP2 meeting.

11.2 WP2/9 meeting during next SG9 meeting

WP2/9 is also planning to meet again during the next SG9 meeting, to be held in March or April 2023 (TBD).

Details will be posted on the ITU-T SG9 website: http://www.itu.int/ITU-T/go/sg9 .

12 Conclusion

The WP2/9 Chairman thanked the delegates for their enthusiastic and active participation in the relevant activities of Working Party 2/9 and each Question during the meeting. Special thanks were expressed to the SG9 Counsellor, Mr Polidori (TSB) and his assistant Ms Hiba Tahawi (TSB) as well as all Associate Rapporteurs, Editors, and contributors for their dedicated and sustained efforts during this meeting. Special congratulations for the newcomers in SG9 who acted as Rapporteurs and Associated Rapporteurs and will continue in these roles in this Study Period, namely: Mr Rakesh DESAI (Q8 Rapporteur and Q3 Associate Rapporteur), Mr Avinash AGARWAL (Q11 Rapporteur) and Ms Qiong YAO (Q8 Associate Rapporteur).

Annex A:
Structure and leadership of Working Party 2/9 (WP2/9)

WP2/9 Management team

– **Chairman:**
Taekyoon KIM (ETRI, Rep. of Korea)

– **Vice-Chairman:**
Eric WANG (Huawei, China)
Pradipta Biswas (Indian Institute of Science, India)

WP2/9 list of Questions and related Rapporteurs

The following table reproduces the current list of WP2/9 Questions and related Rapporteurs.

Question	Title	**Rapporteur**	**Associate Rapporteurs**
Q3/9	AI-enabled enhanced functions over integrated broadband cable network	**Yanbin (Evan) SUN** (Huawei, China)	**Avinash AGARWAL** (Ministry of Communications, Department of Telecommunications, India)
Q5/9	Software components, application programming interfaces (APIs), frameworks and overall software architecture for advanced content distribution services within the scope of Study Group 9	**Haifeng YAN** (Huawei, China)	
Q8/9	The Internet protocol (IP) enabled multimedia applications and services for cable television networks enabled by converged platforms	**Rakesh DESAI** (Ministry of Communications, Department of Telecommunications, India)	**Qiong YAO (Ms)** (NRTA, China)
Q9/9	Requirements, methods, and interfaces of the advanced service platforms to enhance the delivery of audiovisual content, and other multimedia interactive services over integrated broadband cable networks	**Eric WANG** (Huawei, China)	**Soonchoul KIM** (ETRI, Rep. of Korea)
Q11/9	Accessibility to cable systems and services	**Avinash AGARWAL** (Ministry of Communications, Department of Telecommunications, India)	**Zhao Ming** (ABS, China)

Annex B:
List of participants

The list of participants was made available in TD13.

Annex C:
A.1 justification template to describe a proposed new Recommendation (normative) in the work programme

Question:	Q8/9	**Proposed new ITU-T Recommendation**		Online, 6-14 September 2022	
Reference and title:	ITU-T J.mma-req - "Requirements of microservice architecture for audio-visual media in the converged media cloud"				
Base text:	SG9-TD163		**Timing:**	2023	
Editor(s):	Jia Tao (jiatao@abp2003.cn)		**Approval process:**	AAP	

Scope (defines the intent or object of the Recommendation and the aspects covered, thereby indicating the limits of its applicability):

This proposal introduces the background and key technical characteristics of the functional requirements of microservice architecture for audio-visual media in converged media cloud, and puts forward the project proposal of the functional requirements of various architectural aspects.

Summary (provides a brief overview of the purpose and contents of the Recommendation, thus permitting readers to judge its usefulness for their work):

This paper proposes requirements of the technical and business aspect of microservice architecture for audio-visual media in converged media cloud, help accelerating the media industry cloudification progress. There are tremendous complexities related to distributed cloud computing based on microservice technologies, media industry needs effective architecture to govern them and place emphasis on industrial business specialties. Hence a common media microservice architecture (MMA) to mitigate development and operational complexities, and bring forward distinctive industrial business service, is much needed to foster the communication and synergy among our industry.

Relations to ITU-T Recommendations or to other standards (approved or under development):

1. The ITU-T X.785 "Guidelines for defining REST-based managed objects and management interfaces" (approved) standard issued by the SG2 group has well defined the specification of the communication protocol and the definition of the Restful interface. This Recommendation will refer to the relevant content.
2. ITU-T J.1301, J.1302, J.1303 "a cloud-based converged media service to support Internet protocol and broadcast cable television" (approved) published by SG9 group emphasizes the interaction between Cloud-based converged media services (CBCMS) and the cloud. This Recommendation deepens and supplements the service capability, and provides detailed provisions for audiovisual media services according to different hierarchical structures.
3. The "Protocols for microservices based intelligent edge computing" and "Cloud computing – Functional requirements of edge cloud management" (under development) being compiled by the SG11 group well stipulate the intelligent edge computing standards based on microservices. This Recommendation will reference this standard specification in the MMA's infrastructure services layer.
4. The "Cloud Computing – Functional requirements of cloud native Platform as a Service "(under development) being compiled by SG13 group well defines the requirements specification for service registration and service discovery. This Recommendation will refer to the relevant content and further deepen it.
5. The idea of layered architecture is well defined in "Cloud Computing - Functional Requirements of PaaS for Cloud Native Applications"(under development) being compiled by SG13 group. Reference will be made to relevant content in this Recommendation.
6. The requirements for containers are defined in "Cloud Computing – Functional requirements for container" (under development) being compiled by SG13 group. Reference will be made to relevant content in this Recommendation.
7. The "Propose to update eight use cases for Y.cccnp-reqrts" (under development) being compiled by the SG13 group proposes the governance specifications for microservices, as well as the governance of services (Service) Load balancing, fault tolerance, monitoring, tracking, logging and chaos The specification for testing to which this Recommendation applies and conforms.
8. The "Technical specification for artificial intelligence cloud platform – AI model development" (under development) standard being compiled by the SG16 group uses the microservice architecture to build an artificial intelligence cloud platform, which is an application scenario for microservices. This standard will refer to relevant content.

Liaisons with other study groups or with other standards bodies:
ITU-T Study Groups 13, Study Groups 16
Supporting members that are committing to contributing actively to the work item:
Ministry of Industry and Information Technology (MIIT), China

AnnexD:
A.1 justification template to describe a proposed new Recommendation (normative) in the work programme

<table>
<tr><td>Question:</td><td>Q8/9</td><td colspan="2">Proposed new ITU-T Recommendation</td><td colspan="2">Online, 6-14 September 2022</td></tr>
<tr><td>Reference and title:</td><td colspan="5">ITU-T J.mma-spec - “Specification of microservice architecture for audio-visual media in the converged media cloud”</td></tr>
<tr><td>Base text:</td><td colspan="2">SG9-TD164</td><td colspan="2">Timing:</td><td>2023</td></tr>
<tr><td>Editor(s):</td><td colspan="2">Jia Tao
(jiatao@abp2003.cn)</td><td colspan="2">Approval process:</td><td>AAP</td></tr>
<tr><td colspan="6">Scope (defines the intent or object of the Recommendation and the aspects covered, thereby indicating the limits of its applicability):</td></tr>
<tr><td colspan="6">This proposal describes the content and key technical characteristics on the specification of media microservice architecture, and puts forward the project proposal of the specification of various architectural aspects.</td></tr>
<tr><td colspan="6">Summary (provides a brief overview of the purpose and contents of the Recommendation, thus permitting readers to judge its usefulness for their work):</td></tr>
<tr><td colspan="6">This paper proposes specification of microservice architecture for audio-visual media in converged media cloud industry, accelerating the service oriented and cloudification progress. There are tremendous complexities related to distributed cloud computing based on microservice technologies, media industry needs effective architecture to govern them and place emphasis on industrial business specialties. Hence a common Media Microservice Architecture (MMA) to mitigate development and operational complexities, and bring forward distinctive industrial business service. A set of specification on media microservice architecture can greatly foster effectively communication and synergy among our industry.</td></tr>
<tr><td colspan="6">Relations to ITU-T Recommendations or to other standards (approved or under development):</td></tr>
<tr><td colspan="6">1. The ITU-T X.785 "Guidelines for defining REST-based managed objects and management interfaces" (approved) standard issued by the SG2 group has well defined the specification of the communication protocol and the definition of the Restful interface. This Recommendation will refer to the relevant content.
2. ITU-T J.1301, J.1302, J.1303 "a cloud-based converged media service to support Internet protocol and broadcast cable television" (approved) published by SG9 group emphasizes the interaction between Cloud-based converged media services (CBCMS) and the cloud. This Recommendation deepens and supplements the Service capability, and provides detailed provisions for audiovisual media services according to different hierarchical structures.
3. The "Protocols for microservices based intelligent edge computing" and "Cloud computing – Functional requirements of edge cloud management" (under development) being compiled by the SG11 group well stipulate the intelligent edge computing standards based on microservices. This Recommendation will reference this standard specification in the MMA's infrastructure services layer.
4. The "Cloud Computing – Functional requirements of cloud native Platform as a Service"(under development) being compiled by SG13 group well defines the requirements specification for service registration and service discovery. This Recommendation will refer to the relevant content and further deepen it.
5. The idea of layered architecture is well defined in "Cloud Computing - Functional Requirements of PaaS for Cloud Native Applications"(under development) being compiled by SG13 group. Reference will be made to relevant content in this Recommendation.
6. The requirements for containers are defined in "Cloud Computing – Functional requirements for container" (under development) being compiled by SG13 group. Reference will be made to relevant content in this Recommendation.
7. The "Propose to update eight use cases for Y.cccnp-reqrts" (under development) being compiled by the SG13 group proposes the governance specifications for microservices, as well as the governance of services (Service) Load balancing, fault tolerance, monitoring, tracking, logging and chaos The specification for testing to which this Recommendation applies and conforms.
8. The "Technical specification for artificial intelligence cloud platform – AI model development" (under development) standard being compiled by the SG16 group uses the microservice architecture to build an artificial intelligence cloud platform, which is an application scenario for microservices. This standard will refer to relevant content.</td></tr>
<tr><td colspan="6">Liaisons with other study groups or with other standards bodies:</td></tr>
</table>

ITU-T Study Groups 13, Study Groups 16
Supporting members that are committing to contributing actively to the work item:
Ministry of Industry and Information Technology (MIIT), China

Annex E:
A.13 justification template to describe a proposed new work item (non-normative) in the work programme

<table>
<tr><td>Question:</td><td>Q11/9</td><td colspan="2">Proposed new ITU-T Technical Report</td><td colspan="2">Online, 6-14 September 2022</td></tr>
<tr><td>Reference and title:</td><td colspan="5">ITU-T JSTR.LCAP "Technical advances, challenges, and best practices in live captioning”</td></tr>
<tr><td>Base text:</td><td colspan="3">SG9-TD159</td><td>Timing:</td><td>2024</td></tr>
<tr><td>Editor(s):</td><td colspan="3">Avinash Agarwal, Ministry of Communications, India
avinash.70@gov.in
Rajiv Khattar, India
rajivkhattar@gmail.com</td><td>Approval process:</td><td>Agreement</td></tr>
<tr><td colspan="6">Purpose and scope (Define what this document will address and its intent or objectives in order to indicate the limits of its applicability):
The objective of the Technical Report is to facilitate the implementation of live captioning and other visual accessibility features in Television Broadcasting, Cable Networks, Direct to Home (DTH), IPTV, other upcoming platforms, and emerging services such as Virtual Reality (VR). Member states will benefit from the experiences of other states, which the report will include as case studies.
The Technical Report will cover various aspects related to visual accessibility features, live captioning in particular, including recent advances, implementation challenges, and global best practices. The report will also discuss the captioning standards and accuracy measuring metrics, such as Word Error Rate (WER).
The Technical Report aims to provide a roadmap to make accessibility faster, accurate, and simpler using advanced technologies such as AI/ ML.</td></tr>
<tr><td colspan="6">Summary (provides a brief overview of the proposal):
In accordance with the United Nations Convention on the Rights of Persons with Disabilities (UN CRPD) that came into force in May 2008, persons with disabilities also have similar and equal rights and the technology should enable the same. This document elaborates on the technological advancements and compliance framework to address this target group.
Captioning is one of the most accepted accessibility enhancing techniques used in multimedia and television. Live captioning poses challenges in accurate and fast translation of the spoken words. Captioning standards are evolving and so are benchmarking practices.
The proposed Technical Report aims to provide guidance on technical aspects and also share global best practices through case studies. Contributions from member states are encouraged.</td></tr>
<tr><td colspan="6">Relations to ITU-T Recommendations or other documents (approved or under development):
None</td></tr>
<tr><td colspan="6">Liaisons with other study groups or with other standards bodies:
IRG-AVA, ITU-R SG6, ITU-T SG16</td></tr>
<tr><td colspan="6">Supporting members that are committing to contributing actively to the work item:
Ministry of Communications, India</td></tr>
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